

1) UART (Serial)

Point-to-point serial communication

Wires

TX, RX, GND, +5

Tx goes to rx and vice-versa

Best used for

- GPS (UART type)
- Telemetry radio
- RC receiver (SBUS / ELRS)
- Companion computer (Jetson ↔ Pixhawk)

Why

- Simple, widely supported
- Low latency

Limitations

- One device per port
- Noise-sensitive on long wires

2) CAN (DroneCAN)

What it is

- Robust **bus-based** communication

Wires

- **CAN-H, CAN-L, GND, +V**

canH goes to canH and canL to canL

Best used for

- GPS + compass (can type)
- LiDAR / rangefinder

Why

- Very noise-immune (so gps is preferred on can)
- Multiple devices on one bus
- Auto-detect & reliable

Limitations

- Slightly more complex
- Needs proper termination

3) I²C

What it is

- Short-range sensor bus

Wires

- **SDA, SCL, GND, +V**

Scl goes to scl and sda to sda

Best used for

- External compass
- Airspeed sensor
- Barometer
- Optical flow (short wires)

Why

- Very simple
- Multiple sensors on same bus

Limitations

- **Very noise-sensitive**
- Not good for long cables

4) USB

What it is

- High-speed data + power interface

Wires

- **D+, D-, +5V, GND**

Best used for

- Pixhawk ↔ PC (setup, logs)
- Companion computer (debugging)
- Camera / storage (on Jetson)

Why

- High bandwidth
- Plug-and-play

Limitations

- Not reliable in flight
- Can disconnect due to vibration

♦ UART

- Data sent **bit-by-bit sequentially**
- Start bit → data bits → stop bit
- One device talks, the other listens

♦ CAN

- Data sent as **structured frames**
- All devices listen on same bus
- Messages have **IDs & error checking**
- Only highest-priority message wins

♦ I²C

- One **master** (Pixhawk), many **slaves**
- Master asks → slave responds
- Uses addresses to identify devices

♦ USB

- Host (PC/Jetson) controls communication
- High-speed packet-based transfer